

300,000 Streets of Melbourne

School Streets Safety Analysis — Technical Overview

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Problem Statement

Melbourne's 300,000-plus streets include walking routes used by school children every day. Yet no systematic, data-driven assessment of *which* routes are dangerous exists at scale. Our industry partner, Regen Melbourne, needed a reproducible method to identify high-risk school streets, quantify the severity of hazards, and generate costed intervention recommendations — all from open government data.

We focused on three secondary schools in the City of Darebin as a proof-of-concept: **Preston High School**, **Reservoir High School**, and **William Ruthven Secondary College**.

The central research question: *Can data tell us which school streets are putting children at risk — and does disadvantage make it worse?*

Technical Architecture

The system is a fully reproducible, end-to-end Python pipeline. Every output is regenerated by running a single command: `python run_all.py`.

Pipeline Stages

#	Stage	What it does
1	Data Ingestion	Downloads crash records (VicRoads), street network (OSM via Overpass API), air quality (EPA AirWatch), crime rates (CSA), socio-economic indices (ABS SEIFA 2021), and demographics (ABS Census 2021).
2	Feature Engineering	Computes 26 spatial and environmental features per school gate: footpath coverage, crossing count, shade ratio, OSM amenity density, PM2.5, crime rate per 100k. Sentinel codes (777/888/999) are filtered before any calculation.
3	HS Scoring	Ten dedicated scorer modules (<code>src/scoring/hs1.py</code> ... <code>hs10.py</code>) each translate raw features into a 0–10 Healthy Streets score using the framework developed by Dr Lucy Saunders (Transport for London, 2015).
4	Severity Classification	<code>severity.py</code> applies rule-based thresholds: Major if $HS2 < 4.0$, $HS1 < 4.0$, or $HS5 < 3.0$; Moderate if two or more indicators fall below 6.0; otherwise Minor .
5	Recommendations	A rule engine maps each sub-threshold indicator to a prioritised intervention (Critical / High / Medium / Low) with estimated cost range and delivery timeframe.
6	Machine Learning	A Ridge regression model (scikit-learn) is trained on all 26 features using Leave-One-Out cross-validation ($n=3$). Identifies which HS scores can be predicted from open data alone.
7	Scenario Engine	Ten intervention templates are applied to each school using a delta method: the model estimates the change in HS scores, anchored to actual baseline values. Produces 30 pre-computed before/after results.
8	Equity & Crash Analysis	SEIFA disadvantage deciles and ABS demographic data are joined to school catchments. Crash records are filtered to 400 m school-gate buffers and broken down by hour.
9	Visualisation	Nine static charts (Matplotlib) covering safety scores, equity, crash trends, demographics, feature importance, and ML predictions. GIS layers exported as GeoPackage (<code>.gpkg</code>) and QGIS project.
10	Dashboard Export	All results, charts, and scenario data are serialised to JSON and embedded in a static GitHub Pages site (Leaflet.js). Zero backend. Zero installation.

Codebase Structure

- `src/scoring/` — One module per HS indicator (hs1–hs10) plus severity classifier
- `src/features/` — `engineer.py`: all 26 feature computations from OSM, EPA, CSA
- `src/ml/` — `train.py`, `predict.py`, `evaluate.py`: Ridge model lifecycle
- `src/scenarios/` — Intervention delta engine; 10 templates
- `src/recommendations/` — Rule-to-intervention mapping with cost/timeline meta-data
- `src/data/` — Download and cache handlers for each external source
- `src/visualisation/` — Chart builders; one function per chart
- `docs/` — Static dashboard (HTML/CSS/JS);

- deployed to GitHub Pages
- **cache/** — Persistent API response cache; avoids repeated downloads
- **outputs/** — All generated files: CSV, PNG, GeoPackage, HTML, PPTX

Analytical Approach

Healthy Streets Framework

The Healthy Streets framework (Saunders, 2015) provides ten indicators scored 0–10. The minimum “good” threshold is 6.0 per indicator. We selected this framework because it is empirically grounded, council-legible, and has been independently validated in multiple cities. Each indicator maps directly to a data source:

Code	Indicator	Measures	Data source
HS1	Footpath accessibility	Width, continuity, surface quality	Field survey
HS2	Easy to cross	Crossing presence, type, visibility	Field survey
HS3	Shade & shelter	Tree canopy ratio, shelter density	OpenStreetMap
HS4	Rest places	Seating, drinking fountain density	OpenStreetMap
HS5	Not too noisy	Traffic volume, road hierarchy	Field survey
HS6	Active travel	Cycling infrastructure, PT proximity	OSM + field
HS7	Feel safe	Crime rate per 100,000 residents	CSA Victoria
HS8	Things to see & do	POI density within 400 m	OpenStreetMap
HS9	Feel relaxed	School zone presence, traffic calming	Field survey
HS10	Clean air	PM2.5 annual average	EPA AirWatch

Machine Learning Design

Ridge regression was selected for its robustness to correlated features (OSM spatial features are inherently collinear) and its interpretability for a stakeholder audience. With only three schools, Leave-One-Out cross-validation was the only statistically sound evaluation strategy. Mean MAE across all ten indicators was **2.88**.

The key ML insight: **HS7 (feel safe)** and **HS10 (clean air)** are predictable from open data with low error. **HS2 (crossing safety)** is not — because crossing *presence* in OSM does not capture crossing *quality*. This directly informs which indicators require field surveys and which can be automated for a network-wide triage.

Equity Analysis

SEIFA IRSD (Index of Relative Socio-economic Disadvantage) deciles from ABS 2021 were joined to school catchments at SA1 level using population-weighted averaging. The Pearson correlation between SEIFA decile and HS overall score was $r = 0.84$ across the three schools — a strong positive relationship suggesting that disadvantaged catchments systematically receive worse walking environments.

Key Findings

Finding 1 — The Paradox: Preston High School has the highest overall HS score (7.2/10) but is the *only* Major-severity school. HS2 = 0.4: there is no pedestrian crossing at the school gate on a high-traffic arterial road. Nine green scores cannot compensate for one safety-critical failure.

Finding 2 — Crashes are rising and school-timed: Pedestrian and cyclist crashes in Darebin LGA increased 192% between 2021 and 2024 (25 to 73). At William Ruthven SC, 100% of crashes within 400 m of the gate occurred during school arrival or pickup windows — invisible in annual totals, critical when time-filtered.

Finding 3 — Disadvantage predicts danger ($r = 0.84$): Reservoir High School sits in a Decile 4 catchment (moderate-high disadvantage) and has the most safety deficiencies. Preston High is in Decile 7 and scores highest. The pattern is consistent with Giles-Corti et al. (2016): lower-income households have lower car ownership, higher walking dependency, and face worse street conditions.

Finding 4 — The cheapest fix is a crossing: A signalised pedestrian crossing at Preston High (cost: \$80k–\$200k, timeframe: <1 year) would upgrade HS2 from 0.4 to approximately 3.5, moving severity from Major to Moderate. The scenario engine quantifies this return on investment before councils commit capital.

How Findings Were Made Actionable

Raw scores alone do not move councils. We translated every number into a decision-ready artefact:

- **Severity label** (Major / Moderate / Minor) — one word that triggers an urgency response without requiring a stakeholder to read ten indicator values.
- **Recommendations table** (14 rows across 3 schools) — each row includes the specific indicator being addressed, a plain-English description, estimated cost range, and delivery timeframe. Filterable by school and priority in the dashboard.
- **Scenario engine** — answers the question councils actually ask: “What do we get for \$80k?” Thirty pre-computed results are instantly accessible in the browser.
- **Interactive dashboard** (GitHub Pages, zero installation) — five sections covering hero stats, an interactive 4-layer map (crash spots, walk/cycle networks, KDE heatmap, SEIFA catchments), per-school radar charts, the scenario explorer, and a one-click recommendations CSV download.
- **GIS export** — .gpkg and .qgz files for council GIS teams to integrate directly into their existing planning workflows.
- **Scalability argument** — the same pipeline, with no code changes, can be run on all 200+ Melbourne secondary schools. Phase 1 (automated triage on HS7 and HS10) requires zero surveyors.

Technology Stack

Layer	Tools & libraries
Data & pipeline	Python 3.12, pandas, geopandas, OSMnx, requests, openpyxl
ML	scikit-learn (Ridge, cross_val_score), joblib (model persistence)
Spatial	Shapely, Fiona, GDAL, GeoPackage, QGIS
Visualisation	Matplotlib, Seaborn
Dashboard	Leaflet.js, Chart.js, Vanilla JS, GitHub Pages (static)
Presentation	python-pptx (automated PPTX generation)
Document	L ^A T _E X (book, report, this document)

Repository: github.com/s4111078-sidd/300-000-School-Streets • **Dashboard:** s4111078-sidd.github.io/300-000-School-Streets • **Run:** `python run_all.py`